they do not affect the results of subsequent measurements. Therefore, the proposition (*A* **or** *B*) has the same meaning as the proposition (*B* **or** *A*).

1.7 TESTING QUANTUM PROPOSITIONS

Now we come to the quantum world that I described earlier. Let us imagine a situation in which someone (or something) unknown to us has secretly prepared a spin in the $\sigma_z = +1$ state. Our job is to use the apparatus $\mathcal{A}$ to determine whether the proposition (*A* **or** *B*) is true or false. We will try using the procedures outlined above.

We begin by measuring σ_z. Since the unknown agent has set things up, we will discover that $\sigma_z = +1$. It is unnecessary to go on: (*A* **or** *B*) is true. Nevertheless, we could test σ_x just to see what happens. The answer is unpredictable. We randomly find that $\sigma_x = +1$ or $\sigma_x = -1$. But neither of these outcomes affects the truth of proposition (*A* **or** *B*).

But now let's reverse the order of measurement. As before, we'll call the reversed procedure (*B* **or** *A*), and this time we'll measure σ_x first. Because the unknown agent set the spin to +1 along the *z* axis, the measurement of σ_x is random. If it turns out that $\sigma_x = +1$, we are finished: (*B* **or** *A*) is true. But suppose we find the opposite result, $\sigma_x = -1$. The spin is oriented along the $-x$ direction. Let's pause here briefly, to make sure we understand what just happened. As a result of our first measurement, the spin is no longer in its original state $\sigma_z = +1$. It is in a new state, which is either $\sigma_x = +1$ or $\sigma_x = -1$. Please take a moment to let this idea sink in. We cannot overstate its importance.

Now we're ready to test the second half of proposition (*B* **or** *A*). Rotate the apparatus $\mathcal{A}$ to the *z* axis and measure σ_z. According to quantum mechanics, the result will be randomly ±1. This means that there is a 25 percent probability that the experiment produces $\sigma_x = -1$ and $\sigma_z = -1$. In other words, with a probability of ¼, we find that (*B* or *A*) is false; this occurs despite the fact that the hidden agent had originally made sure that $\sigma_z = +1$.